INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Domaine Scientifique de la DOUA
20, Avenue Albert Einstein - 69100 Villeurbanne

Exact and approached optimization

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IDENTIFICATION

CODE : GI-4-S1-EC-OEA
ECTS : 2

HOURS

Cours :	0h
TD :	0h
TP :	28h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	28h
Travail personnel :	12h
Total :	40h

ASSESMENT METHOD

Lab: 100 %

TEACHING AIDS

Slides et Polycopié

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :
Provide the basis of operational research (concepts, methods, models) for industrial engineers.
Enable the students to master linear programming, integer linear programming, graph theory, to model and solve industrial and logistic problems, using optimization sofwares.

CONTENT

The first half of the course alternates between 2-hour lectures and practical sessions, introducing the main concepts of exact optimization and key metaheuristics. It covers a review of linear programming and the simplex method, as well as complexity, integer linear programming, path-based heuristics, and population-based heuristics.

The second half of the course is dedicated to a project. In 2024-25, it focused on a two-tier urban logistics problem, where a truck tour supplies depots, and bicycles then collect goods from the depots to deliver them to customers' homes. A code implementing an algorithmic solution to the problem is provided. Students are then required to replace the solution components with more advanced exact or heuristic methods.

BIBLIOGRAPHY

PRE-REQUISITES

Probability and statistics
Algorithmics
Programming
Integer Linear programming